

Title: Multi – Agent AI Framework for Automated Software Development

Abstract:

Software development involves several stages — requirement gathering, system design, coding, testing, debugging, and documentation — typically handled by different team members, demanding significant time and coordination. This project proposes a Multi-Agent AI Framework for Automated Software Development, where multiple LLM-based agents collaborate to automate different phases of the Software Development Life Cycle (SDLC).

The framework is orchestrated using LangGraph, which manages agent workflows and communication through a central coordinator agent. Each agent is assigned a specific SDLC task — requirement analysis, software design, code generation, testing, or documentation. To balance performance and reliability, the system uses a mixed-model setup: a stronger, more capable model handles the orchestrator/manager agent, where robust reasoning and reliable tool-calling are critical, while LLaMA models served via Groq power the higher-volume, task-specific agents, offering fast inference at no cost. A Flask-based frontend provides the interface for developers to submit requirements and view generated outputs.

By interpreting natural language requirements and generating software artifacts collaboratively, the agents work together to improve efficiency, reduce manual effort, and produce higher-quality software. The system helps developers by accelerating development cycles, catching errors early, and auto-generating documentation — all while relying entirely on free-tier tools and APIs to keep the framework accessible. Overall, the proposed framework demonstrates a faster, more efficient, and more reliable approach to software development through coordinated multi-agent automation.

Datasets:

HuggingFace	https://huggingface.co/datasets/BasitAliiii/sdlc-scenario-based-qa-dataset
Github	https://github.com/GPT-Laboratory/CodePori_v02

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